

INEQUALITIES

Directions for questions 1 to 6: Select the correct alternative from the given choices.

Which of the following is the solution for the inequation $3x + 7 \geq 5x - 9$?

☐ a) $[8, \infty)$

☐ b) $[-8, 8]$

☒ c) $(-\infty, 8]$ (Your answer - ✓)

☐ d) $[0, 8]$

Directions for questions 1 to 6: Select the correct alternative from the given choices.

If $x > 0$, the solution set of the inequations $3x - 8 \leq 7$ and $2x - 9 \leq -x$ is _____.

☐ a) $[0, 5]$

☐ b) $(0, 3]$

☐ c) $[3, 5]$

☐ d) $(1, 3]$

Solution:

$$\text{Given } 3x - 8 \leq 7$$

$$3x \leq 15$$

$$x \leq 5 \rightarrow (1)$$

$$2x - 9 \leq -x$$

$$3x \leq 9$$

$$x \leq 3 \rightarrow (2)$$

$$\text{and } x > 0 \rightarrow (3)$$

$$\therefore \text{ from (1), (2) and (3) } x \in (0, 3]$$

Choice (B)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

If $x > \frac{5}{2}$, the solution set of the inequation $\frac{1}{2x-5} \leq \frac{2}{3}$ is _____.

- ☐ a) $[\frac{13}{4}, \infty)$
- ☒ b) $(\frac{5}{2}, 7)$
- ☐ c) $(\frac{5}{2}, \infty)$
- ☐ d) $(\frac{13}{4}, \infty)$

Solution:

$$\frac{1}{2x-5} - \frac{2}{3} \leq 0$$

$$\frac{3-4x+10}{3(2x-5)} \leq 0$$

$$\Rightarrow \frac{(13-4x)(2x-5)}{3(2x-5)^2} \leq 0$$

$$\Rightarrow (13-4x)(2x-5) \leq 0$$

$$(4x-13)(2x-5) \geq 0$$

$$x \geq \frac{13}{4} \text{ or } x \leq \frac{5}{2}$$

Since $x > \frac{5}{2}$ the solution is $[\frac{13}{4}, \infty)$ Choice (A)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

If $(x + 1)(x - 2)(x + 3) > 0$, x belongs to _____.

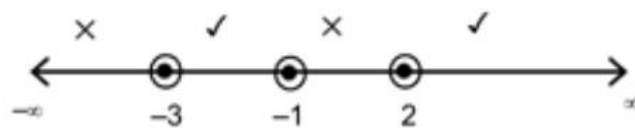
- ☐ a) $(-1, 3)$
- ☐ b) $(-\infty, -3)$
- ☐ c) $(-3, \infty)$
- ☐ d) $(-3, -1) \cup (2, \infty)$

Solution:

$$(x + 1)(x - 2)(x + 3) > 0$$

Critical points are $-3, -1, 2$

Representing these points on the number line we have



When $x = 0$ the given inequation does not hold

$\therefore x \notin (-1, 2)$ and also $x \notin (-\infty, -3)$

$\therefore$ The solution region is $(-3, -1) \cup (2, \infty)$

Choice (D)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

If $2x^2 - x - 6 < 0$, $x \in$ _____.

- ☐ a) $(-\infty, \frac{-3}{2})$
- ☐ b) $(2, \infty)$
- ☐ c) $(\frac{-3}{2}, 2)$
- ☐ d) $(-2, \frac{3}{2})$

Solution:

$$\begin{aligned} 2x^2 - x - 6 &< 0 \\ \Rightarrow (2x + 3)(x - 2) &< 0 \\ x &\in \left(-\frac{3}{2}, 2\right) \end{aligned}$$

Choice (C)

Directions for questions 1 to 6: Select the correct alternative from the given choices.

If $2x^2 - 5x + 18 > 0$, then $x \in$

- ☐ a) ϕ
- ☐ b) $\mathbb{R}$
- ☐ c) $(-3, 5)$
- ☐ d) $(-2, 9)$

Solution:

$$2x^2 - 5x + 18 > 0$$

$$x^2 - \frac{5}{2}x + 9 > 0$$

$$x^2 - 2\frac{5}{4}x + \frac{25}{16} + 9 - \frac{25}{16} > 0$$

$$\left(x - \frac{5}{2}\right)^2 + \frac{119}{16} > 0$$

$$\text{Since } \left(x - \frac{5}{2}\right)^2 > 0$$

$\therefore$ for all values of x the above inequations is true.

$$\Rightarrow x \in \mathbb{R}$$

Choice (B)

Directions for question 7: Type in your answer in the input box provided below the question.

The number of integer solutions of $|x + 3| + |x - 4| < 3$ is

Solution:

We know that the minimum value of $|x - a| + |x - b|$ occurs when $x = \frac{a + b}{2}$.

Here $a = -3$; $b = 4$

$\therefore$ at $x = \frac{-3 + 4}{2} = \frac{1}{2}$, the expression has minimum value.

$$|x + 3| + |x - 4| = \left|\frac{1}{2} + 3\right| + \left|\frac{1}{2} - 4\right| = 7$$

$\therefore$ there is no integer value such that $|x + 3| + |x - 4| < 3$.

Ans : (0)

Directions for question 8: Select the correct alternative from the given choices.

The range of values of x that satisfy the inequality $\frac{1}{|3x-8|} > \frac{3}{5}, x \neq \frac{8}{3}$ is _____.

- ☐ a) $\left(\frac{3}{5}, \frac{8}{3}\right)$
- ☐ b) $\left(\frac{19}{9}, \frac{29}{9}\right)$
- ☐ c) $\left(\frac{-3}{5}, \frac{8}{3}\right)$
- ☐ d) $\left(\frac{-8}{3}, \frac{29}{9}\right)$

Solution:

$$\frac{1}{|3x-8|} > \frac{3}{5}$$

$$\Rightarrow |3x-8| < \frac{5}{3}$$

$$\Rightarrow -\frac{5}{3} < 3x-8 < \frac{5}{3}$$

$$\Rightarrow -\frac{5}{3} + 8 < 3x < \frac{5}{3} + 8$$

$$\Rightarrow \frac{19}{3} < 3x < \frac{29}{3}$$

$$\Rightarrow \frac{19}{9} < x < \frac{29}{9}$$

$$x \in \left(\frac{19}{9}, \frac{29}{9} \right)$$

Choice (B)

Directions for question 9: Type in your answer in the input box provided in the question.
The number of ordered pairs (x, y) that satisfy the equation $3x + 4y = 63$, where $x, y \in \mathbb{N}$ is

Solution:

$$3x + 4y = 63$$
$$\Rightarrow y = \frac{63 - 3x}{4}$$

Possible values of x and their corresponding values of y are listed below

x	1	5	9	13	17
y	15	12	9	6	3

$\Rightarrow$ Number of possible ordered pairs of (x, y) is 5

Ans : (5)

Directions for questions 10 and 11: Type in your answer in the input box provided below the question.

In an obtuse angled triangle, 21 times of one angle is equal to 17 times of another angle. If each angle measures an integral number of degrees, find the least possible measure of any angle in the triangle.

Solution:

Let A, B, C be the angles of the triangle and let $21A = 17B$

$$\Rightarrow A = \frac{17}{21}B$$

Possible values of angles of $\triangle ABC$ are listed below

A°	B°	C°
17	21	142
34	42	104

Only two possibilities. Of them least possible angle is 17° .

Ans : (17)

Directions for questions 10 and 11: Type in your answer in the input box provided below the question.

Two integers greater than 12 are such that the sum of 5 times a number and 6 times another number is 163. Find the difference of the numbers.

Solution:

Let the two positive numbers be x and y

Accordingly $5x + 6y = 163$

$$\Rightarrow y = \frac{163 - 5x}{6}$$

(x, y) can take the values (5, 23), (11, 18), (17, 13), (23, 8), (29, 3).

Since x, y > 12, x = 17, y = 13 is the only possibility.

$\Rightarrow$ Required difference = $17 - 13 = 4$

Ans : (4)

Directions for questions 12 to 14: Select the correct alternative from the given choices.

If the remainder of $\left(\frac{7x}{9}\right) = 7$, and n is any natural number, x can be expressed as ____.

- ☐ a) $9n + 1$
- ☐ b) $9n + 2$
- ☐ c) $7n - 1$
- ☐ d) $7n + 2$

Solution:

Given remainder $\left(\frac{7x}{9}\right) = 7$

The values of x follows an arithmetic progression whose common difference is 9 with first term being 1 which can be expressed as $9n + 1$, $n \in \mathbb{N}$

Choice (A)

If the remainder of $\left(\frac{7x}{9}\right) = 7$, and n is any natural number, x can be expressed as ____.

- ☐ a) $9n + 1$
- ☐ b) $9n + 2$
- ☐ c) $7n - 1$
- ☐ d) $7n + 2$

Solution:

Given remainder $\left(\frac{7x}{9}\right) = 7$

The values of x follows an arithmetic progression whose common difference is 9 with first term being 1 which can be expressed as $9n + 1$, $n \in \mathbb{N}$

Choice (A)

Directions for questions 12 to 14: Select the correct alternative from the given choices.

If

$$\frac{x+2}{x-3}$$

$\leq 0, x \neq 3$ then _____.

- ☐ a) $-2 \leq x \leq 3$
- ☐ b) $-2 \leq x < 3$
- ☐ c) $-3 \leq x \leq 2$
- ☐ d) $-3 < x \leq 2$

Solution:

$$\frac{x+2}{x-3} \leq 0, x \neq 3$$

$$\Rightarrow \frac{(x+2)(x-3)}{(x-3)^2} \leq 0$$

$$\Rightarrow (x+2)(x-3) \leq 0$$

$$\Rightarrow -2 \leq x < 3 \quad (\because x \neq 3)$$

Choice (B)

If

$$\frac{2x-1}{x+2}$$

≥ 0 , $x \neq -2$ then $x \in$ _____.

☐ a) $\left(-2, \frac{1}{2}\right)$

☐ b) $\left[-2, \frac{1}{2}\right]$

☐ c) $[-2, \infty)$

☐ d) $(-\infty, -2) \cup \left[\frac{1}{2}, \infty\right)$

Solution:

$$\begin{aligned}\frac{2x-1}{x+2} &\geq 0 \\ \Rightarrow \frac{(2x-1)(x+2)}{(x+2)^2} &\geq 0 \Rightarrow (2x-1)(x+2) \geq 0 \\ \Rightarrow x < -2 \text{ (}\therefore x \neq -2\text{)} \text{ or } x &\geq \frac{1}{2} \\ \text{The solution set is } (-\infty, -2) \cup &\left[\frac{1}{2}, \infty\right).\end{aligned}$$

Choice (D)

Directions for question 15: Type in your answer in the input box provided below the question.

How many two digit numbers are ten times the sum of their digits?

Directions for question 15: Type in your answer in the input box provided below the question.

How many two digit numbers are ten times the sum of their digits?

Solution:

let the two digit number be ab

$$10a + b = 10(a + b)$$

$$b = 0$$

the numbers possible are 10, 20,, 90

Ans : (9)